

1. Which of the following is a correct Python program to obtain the Python version you are using?

1 point

A.

```
print(__version__)
```

B.

```
import sys
```

```
print(sys.version)
```

C.

```
print(version)
```

D.

```
import sys
```

```
print(sys.__version__)
```

☒ B

☐ A, B, C

☐ B, C, D

☐ A, B

2. What does the following code do?

1 point

```
1 import random
2 def create_dna(n, alphabet='acgt'):
3     return ''.join([random.choice(alphabet) for i in range(n)])
4 dna = create_dna(1000000)
```

☒ Creates a dna variable containing a string of length 1000000, and with the a,c,g,t characters.

☐ Creates a dna variable containing a string of length less than 999999, and with the a,c,g,t characters.

☐ Creates a dna variable containing a string of length 999999, and with the a,c,g,t characters.

☐ Creates a dna variable containing a string of length 999999, containing the 'acgt' substring repeated.

3. The following functions are all supposed to count how many times a certain base (represented as a character variable in Python) appears in a dna sequence (represented as a string variable in Python):

1 point

```
1 def count1(dna, base):
2     i = 0
3     for c in dna:
4         if c == base:
5             i += 1
6     return i
7
8 def count2(dna, base):
9     i = 0
10    for j in range(len(dna)):
11        if dna[j] == base:
12            i += 1
13    return i
14
15 def count3(dna, base):
16     match = [c == base for c in dna]
17     return sum(match)
18
19 def count4(dna, base):
20     return dna.count(base)
21
22 def count5(dna, base):
23     return len([i for i in range(len(dna)) if dna[i] == base])
24
25 def count6(dna, base):
26     return sum(c == base for c in dna)
```

Which of them is correct?

- ☒ All of them are correct.
- ☐ count4, count5 only
- ☐ count2, count3 only
- ☐ count1, count2, count3, and count4 only

4. Which of the correct functions defined in the previous exercise is the fastest?

1 point

Hint. You will need to generate a very large string to test them on, and the function `clock()` from the `time` module to time each function.

- ☒ count4
- ☐ count5
- ☐ count6
- ☐ count1

5. If the `PYTHONPATH` environment variable is set, which of the following directories are searched for modules?

1 point

- A) `PYTHONPATH` directory
- B) current directory
- C) home directory
- D) installation dependent default path

- ☒ A, B, and D
- ☐ A only
- ☐ A and D
- ☐ A, B, and C

6. A student imports a module called `dnautil` in Python using the following command:

1 point

```
import dnautil
```

What does the following call to the `dir` function do?

```
dir(dnautil)
```

- ☒ Lists all the attributes of the `dnautil` module
- ☐ Lists all the functions in the `dnautil` module
- ☐ Lists the `gc` and `has_stop_codon` functions
- ☐ Lists the variables defined in the `dnautil` module

